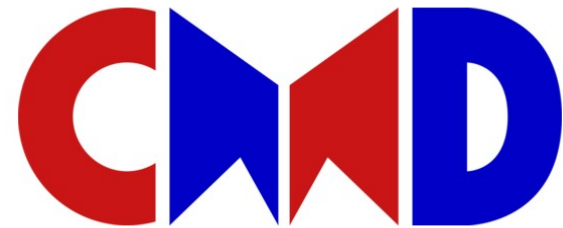

RTC-TOOLS

USER'S MANUAL

Version 1.14i



Real-Time Clock Management for Commodore 64/128

CMD • Ultimate • sd2iec/uIEC • Internet Time

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Project Documentation Edition

About This Edition

RTC-TOOLS V1.14i is a machine-language utility for finding, viewing and synchronizing real-time clocks used with Commodore systems. It combines support for classic CMD RTC hardware with Ultimate-family clocks, RTC-equipped sd2iec/uIEC devices, and optional Internet time obtained through an Ultimate network connection.

***Note:** The implementation source and complete machine-code listing are intentionally not reproduced in this user manual. The Technical Reference explains the system-level behavior without requiring programming knowledge.*

Document Conventions

- Keys to press are shown in capitals, such as RETURN, Y, N, X, + and -.
- Screen messages are reproduced in fixed-width type where useful.
- DEV means an IEC device number. RTC-TOOLS scans IEC device numbers 8 through 30.
- PORT means Commodore control port 1 or control port 2.
- Displayed time means the running date and time shown on the RTC-TOOLS main menu.
- Detected clocks are the RTC devices found by the most recent Search for RTC Devices operation.
- UTC offset means the signed difference between local time and Coordinated Universal Time.

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Before You Begin

- Remove ordinary joysticks before running a full RTC search. Control ports are actively tested for SmartMouse/SmartTrack-family RTC hardware.
- Leave a CMD SmartMouse or SmartTrack connected if you want its RTC detected.
- Make sure all RTC-containing IEC devices are powered, connected and assigned unique device numbers.
- Use a writable program disk/image if you want RTC-TOOLS to create and maintain RTC CONFIG for Internet Time.

- Check the displayed time carefully before using Adopt Displayed Time or Update One Clock.

Section 1

General Information

Introduction and Purpose

A Commodore system can contain several independent real-time clocks. A CMD hard drive, FD-series drive, RAMLink, SmartMouse/SmartTrack, Ultimate-family system, or an RTC-equipped sd2iec/uIEC may each keep its own date and time. RTC-TOOLS provides one consistent interface for finding, comparing and updating those clocks.

- Detect supported RTC devices.
- Maintain a running reference date/time on the main menu.
- View each detected RTC and, when available, its device information.
- Set the displayed date/time manually.
- Write the displayed time to all detected clocks or to one selected clock.
- Use Internet/NTP time through an Ultimate network connection.
- Store a persistent UTC offset in RTC CONFIG.

Supported RTC Families

The supported-device list is alphabetically arranged and includes:

- CMD FD-2000 / FD-4000 / GFD-4000 RTC
- CMD HD-SERIES RTC
- CMD RAMLINK / RAMLINK/XL RTC
- CMD SMARTMOUSE RTC
- CMD SMARTTRACK RTC
- SD2IEC / UIEC RTC
- ULTIMATE 64 / C64 ULTIMATE RTC

Note: A responding IEC device is not automatically counted as an RTC. RTC-TOOLS separately tests whether a valid clock can be read.

PAL and NTSC Operation

RTC-TOOLS detects PAL or NTSC operation at startup. The label PAL or NTSC appears near the lower-right of normal program screens but does not appear on the initial PLEASE WAIT screen.

The detected video standard is also used to keep software timing correct. The main-menu reference clock advances at 50 jiffies per second on PAL and 60 jiffies per second on NTSC. Startup clock-running tests and five-second transient messages use the appropriate timing for the machine.

Important Limits

- The displayed date range is intended for years 2000 through 2099.
- IEC searches cover device numbers 8 through 30.

Appendix B

Messages and Status Text

Message	Meaning
PLEASE WAIT	Startup initialization is in progress.
SCANNING: COMPLETE	The full RTC device search has finished.
RTC CONFIG SAVED.	The UTC offset config write completed.
RTC CONFIG SAVE FAILED.	The config could not be written; a DOS error may be shown underneath.
INTERNET TIME UPDATED.	An Internet/NTP request succeeded and the displayed reference time was updated.
INTERNET TIME UNAVAILABLE.	The Internet/NTP path did not produce a usable time.
TIMEZONE NOT CONFIGURED	No valid persisted UTC offset was loaded.
CLOCK READ FAILED	A previously detected RTC could not be read at that moment.
RTC UPDATE OK	The selected hardware RTC update and verification succeeded.
RTC UPDATE FAILED	The write or verification did not complete successfully.
ADOPTION COMPLETE	Adopt Displayed Time finished processing the detected RTC devices.

Appendix A

Quick Reference

Main Menu

1	List Supported RTC Devices
2	Search for RTC Devices
3	View Detected Clocks
4	Set Date and Time
5	Adopt Displayed Time for All Devices
6	Update One Clock
7	Internet Time
8	Return to BASIC

Internet Time

1	Get Time from Internet
2	Change UTC Offset
3	Return to Main Menu

UTC Offset Editor

+	Increase 15 minutes; hold to repeat
-	Decrease 15 minutes; hold to repeat
RETURN	Accept and save
X	Cancel without saving

Search Preparation

- Remove ordinary joysticks.
- Leave SmartMouse/SmartTrack connected if it should be detected.
- Connect and power all RTC-containing IEC devices.
- Press Y to scan or N to cancel.

- Internet Time requires an Ultimate-family system with Command Interface/network support enabled.
- The configured UTC offset is an effective offset. RTC-TOOLS does not automatically apply daylight-saving-time rules.
- This edition is intended for C64 operation and for a Commodore 128 running in C64 mode.

Section 2

Getting Started

Loading and Starting RTC-TOOLS

RTC-TOOLS is supplied as a normal Commodore PRG with a BASIC entry point. From the device containing the program:

```
LOAD"RTC-TOOLS-V1.14I",8
RUN
```

Note: Substitute the actual device number if the program disk is not device 8.

PLEASE WAIT Screen

Immediately after RUN, RTC-TOOLS displays PLEASE WAIT while initialization takes place. No green divider bars or PAL/NTSC indicator are shown on this screen.

What Happens at Startup

Startup is ordered so a usable time-zone offset exists before Internet time is attempted:

- RTC-TOOLS determines the IEC device from which the program was loaded.
- It looks for the RTC CONFIG sequential file on that device.
- If RTC CONFIG is absent and the disk is writable, RTC-TOOLS creates it with a default offset of UTC +00:00.
- If a valid config is available, Internet/NTP time may be attempted through an Ultimate network interface.
- If Internet time is unavailable, RTC-TOOLS falls back to its normal running-clock discovery.
- The first usable time becomes the reference for the running main-menu display.

Recommended First Session

- Let RTC-TOOLS reach the main menu.
- Choose List Supported RTC Devices to review the supported families.
- Choose Search for RTC Devices and run a complete search.
- Choose View Detected Clocks to compare the hardware clocks.
- Open Internet Time and set the correct UTC offset for your location.
- Use Get Time from Internet if an Ultimate network connection is available.
- Only then use an update option if you intend to change hardware RTCs.

Section 8

Technical Reference

Reference-Time Priority

RTC-TOOLS establishes a usable UTC offset/config before time discovery. It can then attempt Internet/NTP time through an Ultimate network interface. If that path fails, startup falls back to running hardware RTC discovery. The resulting reference value initializes the software clock displayed on the main menu.

Ultimate Command Interface

Ultimate support is accessed directly through the Command Interface rather than by pretending the Ultimate RTC is an IEC disk clock. RTC-TOOLS can read the Ultimate clock, qualify it as running, include it in searches, and update it when firmware policy allows.

sd2iec/uIEC RTC Interface

RTC-equipped sd2iec/uIEC devices expose ISO-format clock commands. RTC-TOOLS identifies compatible firmware, reads time with T-RI and writes time with T-WI followed by an ISO date/time value. The firmware handles the underlying RTC hardware.

Config Record

RTC CONFIG uses a compact eight-byte record:

Offset	Meaning
0-3	ASCII RTCI signature
4	Format version (1)
5-6	Signed UTC offset in minutes, little-endian
7	XOR checksum of bytes 0-6

Search Counts

The completed search keeps separate category counts for CMD IEC devices, CMD Smart devices, Ultimate-family devices and sd2iec/uIEC devices. RTC DEVICES FOUND is the overall detected RTC total.

Clock Write Verification

Clock writes are followed by readback where the device protocol supports it. A small forward time tolerance is allowed because the clock continues to run while the write/read transaction completes.

sd2iec/uIEC RTC Problems

RTC-TOOLS first identifies sd2iec/uIEC firmware and then uses T-RI/T-WI clock commands. Not every sd2iec hardware build includes an RTC. A device without RTC support is not counted as an RTC device.

Section 3 Main Menu and Supported Devices

Main Menu

The bottom prompt is centered and reads **PRESS A KEY <1-8> FOR SELECTION.**

Key	Function
1	LIST SUPPORTED RTC DEVICES
2	SEARCH FOR RTC DEVICES
3	VIEW DETECTED CLOCKS
4	SET DATE AND TIME
5	ADOPT DISPLAYED TIME FOR ALL DEVICES
6	UPDATE ONE CLOCK
7	INTERNET TIME
8	RETURN TO BASIC

List Supported RTC Devices

Option 1 is informational. It displays the RTC families understood by the current program. The list is single-spaced and alphabetically arranged. Press any key to return to the main menu.

The Live Reference Clock

The clock displayed on the main menu is an internal software reference clock. RTC-TOOLS establishes an initial reference value from Internet time or from a running hardware RTC, then advances that displayed time using the C64 jiffy clock.

The main-menu display is not a continuous once-per-second reread of a hardware RTC. RTC-TOOLS periodically resynchronizes from its startup source while appropriate, but a manually overridden displayed time remains under software control.

Note: Because PAL and NTSC use different jiffy rates, RTC-TOOLS adjusts the software-clock rate automatically so one displayed second corresponds to one real second on either video standard.

Section 4

Searching and Viewing RTC Devices

Starting a Search

Choose Option 2, SEARCH FOR RTC DEVICES. Before the scan begins, RTC-TOOLS displays connection warnings. Confirm that ordinary joysticks are removed, SmartMouse/SmartTrack hardware remains connected if desired, and all RTC-containing IEC devices are connected.

The bottom prompt reads <Y> TO SEARCH OR <N> TO CANCEL. Press Y to start or N to cancel. Once Y is pressed, the warning text is removed and the active scanning display moves into the scan area.

Visible Scan Phases

The scanning line is indented and shows the current phase. Live found-device/result lines are also indented one character for consistent alignment. Depending on connected hardware, you will see phases such as:

```
SCANNING: ULTIMATE RTC
SCANNING: CONTROL PORT 1
SCANNING: CONTROL PORT 2
SCANNING: CMD IEC RTC
SCANNING: SD2IEC / UIEC RTC
```

Progress dots are displayed beneath the scanning line. The dot row is reset before the sd2iec/uiEC phase so that progress for the second IEC pass is visible.

Scan Completion and Summary

When the scan is finished, residual progress dots and scan-phase text are cleared and the status becomes:

```
SCANNING: COMPLETE
```

The summary then reports separate counts for CMD IEC devices, CMD Smart devices, Ultimate-family devices, sd2iec/uiEC devices, and the total number of RTC devices found.

View Detected Clocks

Option 3 reads the devices in the current detected list and displays their clock values. The numbered device lines align with the left edge of the DETECTED RTC DEVICES heading. If the list is longer than one screen, RTC-TOOLS continues on additional pages.

A detected device that cannot be read at viewing time is reported with a read failure rather than being silently removed from the list.

Section 7

Troubleshooting

No RTC Devices Are Found

- Confirm ordinary joysticks are removed before the search.
- Confirm SmartMouse/SmartTrack hardware is connected to the port being tested.
- Confirm IEC devices are powered and assigned unique device numbers from 8 through 30.
- Check the IEC cable chain for a device holding the bus.
- Remember that a detected device may legitimately report NO RTC if no clock is installed.

Drive Error Light While Using Internet Time

RTC-TOOLS uses the program disk/image to load and store RTC CONFIG. The image must be writable if the config has to be created or changed. If a disk error occurs, the Internet Time screen can display the DOS error channel beneath the menu after a blank line.

Common examples include FILE NOT FOUND on a first probe, WRITE PROTECT ON for a read-only image, or a disk-full/file-system error. Normal first-run file absence is handled by creating RTC CONFIG with UTC +00:00.

RTC CONFIG Does Not Retain the Offset

- Make sure you press RETURN in the UTC editor; X deliberately cancels.
- Wait for RTC CONFIG SAVED before leaving the editor.
- Use the same writable disk image when testing persistence. A newly downloaded D64 or D1M does not inherit RTC CONFIG from an older image.
- Exit and re-enter Internet Time to force the persisted config to be reread.

Internet Time Is Unavailable

- Confirm the Ultimate Command Interface is enabled.
- Confirm the Ultimate has a working network connection and DNS access.
- Confirm the configured UTC offset is reasonable.
- If the time server does not respond, the normal NTP reply wait ends after approximately 2.0 seconds on NTSC or 2.4 seconds on PAL.
 - If Internet time fails, verify that normal hardware-clock startup still works; the network feature is intended to fail safely.

Ultimate RTC Cannot Be Updated

The Ultimate firmware can prohibit date/time changes. Enable the appropriate Ultimate date/time setting before retrying. RTC-TOOLS treats a prohibited write as a failure.

- + increases the UTC offset by 15 minutes. Holding + repeats the 15-minute adjustment after a short initial delay; releasing the key stops the repeat.
- - decreases the UTC offset by 15 minutes. Holding - repeats the 15-minute adjustment after a short initial delay; releasing the key stops the repeat.
- RETURN accepts the value and immediately writes it to RTC CONFIG.
- X cancels and restores the value that was present when the editor was opened.

When + or - is pressed or held, only the UTC value itself is repainted. The rest of the screen is left unchanged. A new repeat sequence requires a new + or - keypress after the previous key has been released.

Saving and Status Messages

When RETURN accepts a changed offset, RTC-TOOLS briefly displays SAVING RTC CONFIG. A successful save is followed by RTC CONFIG SAVED. A failed save is followed by RTC CONFIG SAVE FAILED and the actual drive/DOS error status when available.

Internet Time status and error messages are centered in the lower status area. Completed messages remain visible for approximately five real seconds and then clear automatically. The timeout is PAL/NTSC aware.

Internet/NTP Synchronization

Choose GET TIME FROM INTERNET to manually request Internet time. RTC-TOOLS contacts an NTP server through the Ultimate network interface, obtains UTC, applies the configured offset, and updates the displayed reference time if the request succeeds. If the NTP server does not answer, the normal reply wait times out after about 2.0 seconds on NTSC or 2.4 seconds on PAL.

Internet/NTP synchronization is also eligible during startup. If the network path is unavailable, RTC-TOOLS continues into its normal hardware-clock discovery instead of blocking startup indefinitely.

Note: The configured offset is the effective UTC offset. If daylight-saving time changes in your location, update the offset manually.

SEQ File Behavior

RTC CONFIG is deliberately a SEQ file rather than a REL file so it remains broadly compatible with Commodore drives, CMD devices, VICE disk images and sd2iec/uIEC environments. The record contains a small signature, format version, signed offset and checksum.

Section 5 Setting and Updating Clocks

Set Date and Time

Option 4 changes the displayed reference date/time. It does not immediately write any hardware RTC. Use the adjustment controls or typed-entry mode to set a correct working value before performing clock updates.

- + increases the selected field. Holding + repeats the adjustment until the key is released.
- - decreases the selected field. Holding - repeats the adjustment until the key is released.
- RETURN accepts the displayed date/time as indicated on screen.
- T enters typed-entry mode for day, month, date, year, hour, minute, second and AM/PM values.
- X returns to the previous menu without accepting the displayed changes or updating an RTC.
 - The screen identifies this control as <X> RETURN TO PREVIOUS MENU.
 - Invalid calendar values are rejected.

Adopt Displayed Time for All Devices

Option 5 writes the displayed reference time to every RTC in the current detected list. Device/result lines are indented one character for consistent alignment. Use it only after a fresh search and after checking the displayed time carefully.

Each device is written and read back. RTC-TOOLS allows a small time window during verification because a real RTC may advance while the transaction is being checked. When the operation finishes, the centered completion message ADOPTION COMPLETE is displayed.

Update One Clock

Option 6 lists detected RTCs and lets you update one selected device. The bottom prompt reads <0> FOR MAIN MENU OR SELECT RTC NUMBER. Enter the RTC number to update, or enter 0 and press RETURN to return to the main menu without updating a clock. After an update, the selected-device line is shown above RTC UPDATE OK or RTC UPDATE FAILED. This is useful for testing a repaired clock, synchronizing only one device, or avoiding changes to other independent RTCs.

Supported Update Paths

- CMD IEC RTCs use the established CMD clock protocol.
- SmartMouse/SmartTrack-family RTCs use the control-port RTC protocol.
- Ultimate-family RTCs use the Ultimate Command Interface SET_TIME function when allowed by Ultimate configuration.

- sd2iec/uIEC RTCs use the firmware T-WI ISO-format time command and are read back with T-RI.

Note: *Ultimate firmware can prohibit date setting. In that case RTC-TOOLS reports the update as failed rather than claiming success.*

Section 6

Internet Time and RTC CONFIG

Overview

RTC-TOOLS supports optional Internet time. An Ultimate-family system can provide network access through the Ultimate Command Interface. RTC-TOOLS can request a base NTP time, apply the configured UTC offset, and use the resulting local date/time as its reference clock.

Internet Time Menu

The Internet Time menu contains three choices: GET TIME FROM INTERNET, CHANGE UTC OFFSET, and RETURN TO MAIN MENU. The three option numbers are aligned vertically, and the centered bottom prompt reads PRESS A KEY <1-3> FOR SELECTION.

The CURRENT UTC OFFSET line is centered and shows the effective time-zone offset that will be applied to UTC obtained from the Internet. The UTC value determines the local time zone used by RTC-TOOLS. Set it correctly for your location before synchronizing.

Examples: Mountain Standard Time uses UTC -07:00; Mountain Daylight Time uses UTC -06:00.

When waiting for an NTP time-server reply, RTC-TOOLS allows 120 video jiffies before treating the request as unanswered. This is approximately 2.0 seconds on NTSC systems and 2.4 seconds on PAL systems. Lower-level Ultimate interface operations have their own safeguards, so a stalled interface can take somewhat longer than the normal NTP-reply timeout.

Key	Function
1	GET TIME FROM INTERNET
2	CHANGE UTC OFFSET
3	RETURN TO MAIN MENU

RTC CONFIG

The time-zone setting is stored in a small sequential file named RTC CONFIG on the same IEC device from which RTC-TOOLS was loaded. Startup checks this file before Internet/NTP access.

If the file does not exist and the disk/image is writable, RTC-TOOLS creates a valid config with UTC +00:00. A missing config therefore does not prevent Internet Time from having a usable offset.

Changing the UTC Offset

Choose CHANGE UTC OFFSET. The offset changes in 15-minute increments so half-hour and quarter-hour time zones are supported. The editor displays USE <+> OR <-> TO CHANGE BY 15 MINUTES and <RETURN> ACCEPT <X> CANCEL.